

CGBSE CLASS 10 MATHEMATICS CURRICULUM

Academic Session 2026–27

Ingestion-Ready Structured Version

DOCUMENT METADATA

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- Question Paper Structure
- Detailed Syllabus
- Project Work Scheme

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theory_marks: 75
project_marks: 25

theory_exam_duration: "3 Hours"
project_exam_duration: "2 Hours"

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theory_periods: 200
project_periods: 20

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primary: Hindi

supporting_terms: English

curriculum_level: Secondary

curriculum_status: Official Academic Curriculum

CHUNK 1: CURRICULUM OVERVIEW

chunk_id: MATH10_CURRICULUM_OVERVIEW

chunk_type: curriculum_overview

section: General Information

Class 10 Mathematics Curriculum Overview

Subject: गणित

English Name: Mathematics

Subject Code: 100

Class: 10

Academic Session: 2026–27

Theory Marks: 75

Project Marks: 25

Total Marks: 100

Theory Examination Duration: 3 Hours

Project Examination Duration: 2 Hours

Total Teaching Periods: 220

Theory Teaching Periods: 200

Project Periods: 20

The Class 10 Mathematics curriculum consists of eight major units:

1. बीजगणित — Algebra
2. निर्देशांक ज्यामिति — Coordinate Geometry
3. वाणिज्य गणित — Commercial Mathematics
4. त्रिकोणमिति — Trigonometry

5. ज्यामिति — Geometry
6. गणितीय कथनों की जाँच — Proof of Mathematical Statements
7. क्षेत्रमिति — Mensuration
8. सांख्यिकी — Statistics

The curriculum is designed to develop:

- Mathematical reasoning
- Numerical accuracy
- Algebraic thinking
- Geometrical understanding
- Graph interpretation
- Problem-solving skills
- Logical proof-writing
- Real-life mathematical application
- Data interpretation
- Financial calculation skills

CHUNK 2: CURRICULUM STRUCTURE

chunk_id: MATH10_CURRICULUM_STRUCTURE

chunk_type: curriculum_structure

section: Appendix A

total_theory_marks: 75

total_project_marks: 25

total_marks: 100

Unit-Wise Marks and Period Distribution

Unit Number	Unit Name	English Name	Marks	Periods
1	बीजगणित	Algebra	20	60
2	निर्देशांक ज्यामिति	Coordinate Geometry	8	15
3	वाणिज्य गणित	Commercial Mathematics	7	15
4	त्रिकोणमिति	Trigonometry	10	30
5	ज्यामिति	Geometry	12	30

6	गणितीय कथनों की जाँच	Proof of Mathematical Statements	4	10
7	क्षेत्रमिति	Mensuration	8	25
8	सांख्यिकी	Statistics	6	15
	Theory Total		75	200
	प्रायोजना कार्य	Project Work	25	20
	Grand Total		100	220

CHUNK 3: UNIT-WISE TOPIC SUMMARY

chunk_id: MATH10_UNIT_TOPIC_SUMMARY

chunk_type: syllabus_summary

section: Curriculum Structure

Unit 1: बीजगणित

Topics:

- बहुपद
- दो चरों का रैखिक समीकरण
- एक चर का द्विघात समीकरण
- समांतर श्रेणी
- अनुपात एवं समानुपात

Marks: 20

Periods: 60

Unit 2: निर्देशांक ज्यामिति

Topics:

- निर्देशांक ज्यामिति
- आलेख

Marks: 8

Periods: 15

Unit 3: वाणिज्य गणित

Topic:

1. बैंकिंग एवं कराधान

Marks: 7

Periods: 15

Unit 4: त्रिकोणमिति

Topics:

1. त्रिकोणमितीय समीकरण एवं सर्वसमिकाएँ
2. ऊँचाई एवं दूरी

Marks: 10

Periods: 30

Unit 5: ज्यामिति

Topics:

1. ज्यामितीय आकृतियों में समरूपता
2. वृत्त एवं स्पर्श रेखाएँ
3. ज्यामितीय रचनाएँ

Marks: 12

Periods: 30

Unit 6: गणितीय कथनों की जाँच

Topic:

1. गणितीय कथनों की जाँच

Marks: 4

Periods: 10

Unit 7: क्षेत्रमिति

Topic:

1. ठोस आकृतियों का पृष्ठीय क्षेत्रफल एवं आयतन

Marks: 8

Periods: 25

Unit 8: सांख्यिकी

Topic:

1. आँकड़ों का विश्लेषण

Marks: 6

Periods: 15

CHUNK 4: UNIT 1 — बीजगणित

chunk_id: MATH10_UNIT1_ALGEBRA

chunk_type: subject_unit

section: Detailed Syllabus

unit_number: 1

unit_name: बीजगणित

unit_english_name: Algebra

marks: 20

periods: 60

topic_count: 5

Unit Overview

बीजगणित इकाई विद्यार्थियों को बहुपदों, रैखिक समीकरणों, द्विघात समीकरणों, समांतर श्रेणी तथा अनुपात-समानुपात से संबंधित संकल्पनाओं और समस्याओं को समझने तथा हल करने में सक्षम बनाती है।

The Algebra unit develops competency in:

- Polynomial operations
 - Factorization
 - Simultaneous linear equations
 - Quadratic equations
 - Arithmetic progressions
 - Ratio and proportion
 - Algebraic modelling
 - Problem-solving using equations
-

CHUNK 5: बहुपद

chunk_id: MATH10_UNIT1_TOPIC1_POLYNOMIALS

chunk_type: mathematics_topic

section: Algebra

unit_number: 1

topic_number: 1

topic_name: बहुपद

topic_english_name: Polynomials

marks_group: Algebra

Topic: बहुपद

The topic covers:

- बहुपदों का भाग
 - शेषफल प्रमेय
 - गुणनखंड प्रमेय
 - बहुपदों का गुणनखंड करना
 - मध्यपद को तोड़कर गुणनखंड करना
 - द्विघाती बहुपद के मान
 - द्विघाती बहुपद के शून्यक
 - गुणकों और शून्यकों के बीच संबंध
-

Detailed Concepts

Polynomial Division

विद्यार्थी एक बहुपद को दूसरे बहुपद से विभाजित करना सीखेंगे।

General division relation:

$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

Remainder Theorem

If a polynomial $(p(x))$ is divided by $(x-a)$, then the remainder is $(p(a))$.

Hindi Concept:

यदि किसी बहुपद $(p(x))$ को $(x-a)$ से विभाजित किया जाता है, तो शेषफल $(p(a))$ होता है।

Factor Theorem

If:

$$p(a) = 0$$

then:

$$x - a$$

is a factor of $(p(x))$.

Hindi Concept:

यदि $(p(a)=0)$, तो $(x-a)$, बहुपद $(p(x))$ का एक गुणनखंड होगा।

Factorization of Quadratic Polynomial

Standard form:

$$ax^2 + bx + c$$

The curriculum includes factorization by splitting the middle term.

Example form:

$$ax^2 + bx + c$$

The middle term (bx) is split into two suitable terms so that grouping can be applied.

Zeros of a Quadratic Polynomial

For a quadratic polynomial:

$$ax^2 + bx + c$$

with zeros (α) and (β):

$$\alpha + \beta = -b/a$$

$$\alpha\beta = c/a$$

Learning Outcomes

Students should be able to:

- Divide one polynomial by another.
 - Apply the remainder theorem.
 - Apply the factor theorem.
 - Factorize quadratic polynomials.
 - Find the zeros of a quadratic polynomial.
 - Verify the relation between zeros and coefficients.
 - Form polynomial expressions from given conditions.
-

Retrieval Keywords

keywords:

- बहुपद
- polynomial
- polynomial division
- शेषफल प्रमेय
- remainder theorem
- गुणनखंड प्रमेय
- factor theorem
- द्विघाती बहुपद
- quadratic polynomial
- शून्यक
- zeros of polynomial
- गुणांक
- coefficients

- मध्यपद विभाजन
 - splitting the middle term
-

CHUNK 6: दो चरों का रैखिक समीकरण

chunk_id: MATH10_UNIT1_TOPIC2_LINEAR_EQUATIONS

chunk_type: mathematics_topic

section: Algebra

unit_number: 1

topic_number: 2

topic_name: दो चरों का रैखिक समीकरण

topic_english_name: Linear Equations in Two Variables

Topic Overview

This topic covers a pair of linear equations in two variables and methods used to determine their solutions.

General form:

$$a_1x + b_1y + c_1 = 0$$

$$a_2x + b_2y + c_2 = 0$$

Detailed Concepts

Formation of Equations from Statements

विद्यार्थी कथनों और वास्तविक जीवन की परिस्थितियों से दो चरों वाले रैखिक समीकरण बनाएँगे।

Examples may involve:

- Age problems
 - Cost and quantity
 - Distance and speed
 - Number-based problems
 - Ratio-based situations
-

Graphical Method

The solution of two linear equations can be obtained by plotting both equations on a coordinate plane.

Possible cases:

1. Intersecting lines — one unique solution
 2. Parallel lines — no solution
 3. Coincident lines — infinitely many solutions
-

Elimination Method

One variable is eliminated by making its coefficients equal and then adding or subtracting the equations.

Hindi Term:

विलोपन विधि

Substitution Method

One variable is expressed in terms of another and substituted into the second equation.

Hindi Term:

प्रतिस्थापन विधि

Nature of Solutions

For equations:

$$a_1x + b_1y + c_1 = 0$$

$$a_2x + b_2y + c_2 = 0$$

Unique Solution

$$a_1/a_2 \neq b_1/b_2$$

No Solution

$$a_1/a_2 = b_1/b_2 \neq c_1/c_2$$

Infinitely Many Solutions

$$a_1/a_2 = b_1/b_2 = c_1/c_2$$

Unknown Coefficients

Students may be asked to determine unknown coefficients based on the type of solution of a pair of equations.

Learning Outcomes

Students should be able to:

- Form equations from verbal statements.
 - Solve simultaneous equations graphically.
 - Solve equations using elimination.
 - Solve equations using substitution.
 - Determine whether equations have one, no, or infinitely many solutions.
 - Find unknown coefficients using solution conditions.
 - create word problems from equations.
-

Retrieval Keywords

keywords:

- दो चरों का रैखिक समीकरण
 - linear equations in two variables
 - युगपद समीकरण
 - simultaneous equations
 - आलेखी विधि
 - graphical method
 - विलोपन विधि
 - elimination method
 - प्रतिस्थापन विधि
 - substitution method
 - एकमात्र हल
 - no solution
 - infinite solutions
-

CHUNK 7: एक चर का द्विघात समीकरण

chunk_id: MATH10_UNIT1_TOPIC3_QUADRATIC_EQUATIONS

chunk_type: mathematics_topic
section: Algebra
unit_number: 1
topic_number: 3
topic_name: एक चर का द्विघात समीकरण
topic_english_name: Quadratic Equations in One Variable

Topic Overview

A quadratic equation in one variable is generally written as:

$$ax^2 + bx + c = 0$$

where:

$$a \neq 0$$

Detailed Concepts

The curriculum includes:

- द्विघात समीकरण की पहचान
- द्विघात समीकरण के मूल
- मूलों की जाँच
- गुणनखंड विधि
- पूर्ण वर्ग विधि
- द्विघात सूत्र
- विभेदक
- मूलों की प्रकृति
- अज्ञात गुणांक
- मूलों और गुणांकों के बीच संबंध
- दिए गए मूलों से द्विघात समीकरण बनाना

Factorization Method

The quadratic expression is factorized and each factor is equated to zero.

Completing the Square

The equation is transformed into a perfect-square form.

Hindi Term:

पूर्ण वर्ग बनाकर हल करना

Quadratic Formula

For:

$$ax^2 + bx + c = 0$$

the roots are:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant

The discriminant is:

$$D = b^2 - 4ac$$

Nature of roots:

$D > 0 \rightarrow$ Two distinct real roots

$D = 0 \rightarrow$ Two equal real roots

$D < 0 \rightarrow$ No real roots

Relation Between Roots and Coefficients

If the roots are (α) and (β) :

$$\alpha + \beta = -b/a$$

$$\alpha\beta = c/a$$

Formation of Quadratic Equation from Roots

If the roots are (α) and (β) , then the equation is:

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

Learning Outcomes

Students should be able to:

- Identify quadratic equations.
 - Find roots using factorization.
 - Solve equations by completing the square.
 - Apply the quadratic formula.
 - Calculate the discriminant.
 - Determine the nature of roots.
 - Find unknown coefficients.
 - verify roots of an equation.
 - Form a quadratic equation from given roots.
-

Retrieval Keywords

keywords:

- द्विघात समीकरण
 - quadratic equation
 - quadratic formula
 - द्विघात सूत्र
 - विभेदक
 - discriminant
 - मूलों की प्रकृति
 - nature of roots
 - गुणनखंड विधि
 - completing the square
 - roots and coefficients
-

CHUNK 8: समांतर श्रेणी

chunk_id: MATH10_UNIT1_TOPIC4_ARITHMETIC_PROGRESSION

chunk_type: mathematics_topic

section: Algebra

unit_number: 1

topic_number: 4

topic_name: समांतर श्रेणी

topic_english_name: Arithmetic Progression

Topic Overview

A sequence is an arithmetic progression when the difference between consecutive terms is constant.

Standard form:

$a, a+d, a+2d, a+3d, \dots$

where:

a = first term

d = common difference

Detailed Concepts

The curriculum includes:

- समांतर श्रेणी की पहचान
 - प्रथम पद
 - सार्व अंतर
 - समांतर श्रेणी का n वाँ पद
 - दो राशियों का समांतर माध्य
 - दो राशियों के बीच समांतर श्रेणी बनाना
 - प्रथम n पदों का योग
-

nth Term

The n th term of an arithmetic progression is:

$$a_n = a + (n - 1)d$$

Arithmetic Mean

The arithmetic mean between two numbers (a) and (b) is:

$$A = (a + b) / 2$$

Sum of First n Terms

$$S_n = n/2 [2a + (n - 1)d]$$

Alternative form:

$$S_n = n/2 (a + l)$$

where (l) is the last term.

Learning Outcomes

Students should be able to:

- Identify an arithmetic progression.
 - Determine the first term and common difference.
 - Find the nth term.
 - Insert arithmetic means between two quantities.
 - Construct an arithmetic progression.
 - Calculate the sum of the first n terms.
 - Apply arithmetic progressions to real-life problems.
-

Retrieval Keywords

keywords:

- समांतर श्रेणी
 - arithmetic progression
 - AP
 - सार्व अंतर
 - common difference
 - nth term
 - nवाँ पद
 - arithmetic mean
 - समांतर माध्य
 - sum of AP
-

CHUNK 9: अनुपात एवं समानुपात

chunk_id: MATH10_UNIT1_TOPIC5_RATIO_PROPORTION

chunk_type: mathematics_topic

section: Algebra

unit_number: 1
topic_number: 5
topic_name: अनुपात एवं समानुपात
topic_english_name: Ratio and Proportion

Topic Overview

This topic focuses on comparison of quantities using ratios and equality between ratios using proportion.

Detailed Concepts

The curriculum includes:

- अनुपात
 - अनुपात का व्यावहारिक उपयोग
 - किसी राशि को दो या अधिक भागों में बाँटना
 - किसी दिए गए अनुपात में राशि का विभाजन
 - समानुपात
 - चतुर्थानुपाती
 - मध्यानुपाती
 - तृतीयानुपाती
 - सतत अनुपात
 - K-नियम
 - व्यक्तमानुपात
-

Ratio

A ratio compares two quantities of the same kind.

Example:

a : b

Proportion

If two ratios are equal:

$$a/b = c/d$$

then:

$$a : b = c : d$$

and:

$$ad = bc$$

Fourth Proportional

If:

$$a : b = c : d$$

then (d) is the fourth proportional to (a), (b), and (c).

Mean Proportional

If:

$$a : x = x : b$$

then:

$$x^2 = ab$$

and:

$$x = \sqrt{ab}$$

Third Proportional

If:

$$a : b = b : c$$

then (c) is the third proportional to (a) and (b).

Learning Outcomes

Students should be able to:

- Compare quantities using ratios.
 - Divide quantities in a given ratio.
 - Solve proportion-based problems.
 - Find the third proportional.
 - Find the fourth proportional.
 - Calculate the mean proportional.
 - Apply ratio and proportion to practical situations.
 - Use proportional reasoning in mathematical problems.
-

Retrieval Keywords

keywords:

- अनुपात
 - ratio
 - समानुपात
 - proportion
 - चतुर्थानुपाती
 - fourth proportional
 - मध्यानुपाती
 - mean proportional
 - तृतीयानुपाती
 - third proportional
 - सतत अनुपात
 - K नियम
-

CHUNK 10: UNIT 2 — निर्देशांक ज्यामिति

chunk_id: MATH10_UNIT2_COORDINATE_GEOMETRY

chunk_type: subject_unit

section: Detailed Syllabus

unit_number: 2

unit_name: निर्देशांक ज्यामिति

unit_english_name: Coordinate Geometry

marks: 8

periods: 15

topic_count: 2

Unit Overview

निर्देशांक ज्यामिति इकाई में निर्देशांक समतल, बिंदुओं का निरूपण, दूरी, ढाल, अंतःखंड, रेखा का समीकरण और आलेखों की व्याख्या शामिल है।

The unit develops:

- Spatial understanding
- Graphical representation
- Coordinate-based calculation
- Interpretation of relationships
- Analytical geometry skills

CHUNK 11: निर्देशांक ज्यामिति

chunk_id: MATH10_UNIT2_TOPIC1_COORDINATE_GEOMETRY

chunk_type: mathematics_topic

section: Coordinate Geometry

unit_number: 2

topic_number: 1

topic_name: निर्देशांक ज्यामिति

topic_english_name: Coordinate Geometry

Detailed Concepts

The curriculum includes:

- निर्देशांक ज्यामिति का परिचय
- निर्देशांक समतल
- किसी बिंदु का प्रदर्शन
- दो बिंदुओं के बीच दूरी
- रेखा की ढाल
- रेखा की प्रवणता
- अक्षों पर रेखा का अंतःखंड
- रेखा का समीकरण

Coordinate Plane

A point is represented as:

(x, y)

where:

x = abscissa

y = ordinate

Distance Between Two Points

For points:

P(x₁, y₁)

Q(x₂, y₂)

the distance is:

$$PQ = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$$

Slope of a Line

The slope of a line passing through two points is:

$$m = (y_2 - y_1) / (x_2 - x_1)$$

Intercepts

The points where a line cuts the coordinate axes are called intercepts.

Hindi Term:

अंतःखंड

Equation of a Line

Common line equation forms may include:

$$y = mx + c$$

and:

$$ax + by + c = 0$$

Learning Outcomes

Students should be able to:

- Plot points on the coordinate plane.
 - Identify quadrants.
 - Calculate the distance between two points.
 - Find the slope of a line.
 - Interpret x-intercepts and y-intercepts.
 - Form and identify equations of straight lines.
-

Retrieval Keywords

keywords:

- निर्देशांक ज्यामिति
 - coordinate geometry
 - coordinate plane
 - निर्देशांक समतल
 - distance formula
 - दूरी सूत्र
 - slope
 - ढाल
 - प्रवणता
 - intercept
 - अंतःखंड
 - equation of line
-

CHUNK 12: आलेख

chunk_id: MATH10_UNIT2_TOPIC2_GRAPHs

chunk_type: mathematics_topic

section: Coordinate Geometry

unit_number: 2

topic_number: 2

topic_name: आलेख

topic_english_name: Graph

Topic Overview

This topic focuses on graphical representation and interpretation of relationships between two quantities.

Detailed Concepts

The curriculum includes:

- दो राशियों के बीच संबंध को आलेख पर देखना
 - दो राशियों के मध्य संबंध को आलेख पर दर्शाना
 - विभिन्न परिस्थितियों में बने आलेखों को पढ़ना
 - आलेखों से निष्कर्ष निकालना
-

Graph Interpretation Skills

Students may interpret graphs related to:

- Distance and time
 - Speed and time
 - Cost and quantity
 - Temperature and time
 - Growth and decline
 - Direct variation
 - Real-life data
-

Learning Outcomes

Students should be able to:

- Plot relationships between two quantities.
 - Read values from graphs.
 - Interpret trends.
 - Compare two graphical relationships.
 - Identify increasing and decreasing patterns.
 - Draw conclusions from graphical information.
 - Represent real-life situations graphically.
-

Retrieval Keywords

keywords:

- आलेख
 - graph
 - graphical representation
 - graph interpretation
 - coordinate graph
 - data graph
 - relationship between variables
 - graphical analysis
-

CHUNK 13: UNIT 3 — वाणिज्य गणित

chunk_id: MATH10_UNIT3_COMMERCIAL_MATHEMATICS

chunk_type: subject_unit

section: Detailed Syllabus

unit_number: 3

unit_name: वाणिज्य गणित

unit_english_name: Commercial Mathematics

marks: 7

periods: 15

topic_count: 1

Unit Overview

वाणिज्य गणित इकाई में बैंकिंग, जमा खातों पर ब्याज तथा आयकर की मूलभूत गणना शामिल है।

The unit connects mathematics with:

- Banking
 - Savings
 - Interest calculation
 - Financial planning
 - Taxation
 - Real-life commercial transactions
-

CHUNK 14: बैंकिंग एवं कराधान

chunk_id: MATH10_UNIT3_TOPIC1_BANKING_TAXATION

chunk_type: mathematics_topic

section: Commercial Mathematics

unit_number: 3

topic_number: 1
topic_name: बैंकिंग एवं कराधान
topic_english_name: Banking and Taxation

Detailed Concepts

The curriculum includes:

- बैंकिंग का परिचय
 - आवर्ती जमा खाता
 - आवर्ती जमा खाते पर ब्याज
 - सावधि जमा खाता
 - सावधि जमा खाते पर ब्याज
 - आयकर का परिचय
 - आयकर की गणना
-

Recurring Deposit

An account in which a fixed amount is deposited regularly for a specified period.

Hindi Term:

आवर्ती जमा खाता

Students learn to calculate interest and maturity amount based on the prescribed formula and deposit period.

Fixed Deposit

A lump-sum amount deposited in a bank for a fixed duration at a specified interest rate.

Hindi Term:

सावधि जमा खाता

Students calculate:

- Principal
 - Interest
 - Maturity amount
 - Rate
 - Time period
-

Income Tax

Income tax is a direct tax charged on taxable income according to prescribed rules and tax slabs.

Hindi Term:

आयकर

Students learn basic income-tax calculation using given taxable income and applicable rates.

Learning Outcomes

Students should be able to:

- Explain basic banking concepts.
 - Calculate recurring-deposit interest.
 - Calculate fixed-deposit interest.
 - Determine maturity amounts.
 - Understand the concept of taxable income.
 - Calculate income tax using provided rates.
 - Apply commercial mathematics to practical financial situations.
-

Retrieval Keywords

keywords:

- वाणिज्य गणित
 - commercial mathematics
 - बैंकिंग
 - banking
 - आवर्ती जमा
 - recurring deposit
 - सावधि जमा
 - fixed deposit
 - ब्याज
 - interest
 - आयकर
 - income tax
 - maturity amount
-

PART 1 COMPLETION METADATA

part_number: 1

part_title: Curriculum Metadata and Units 1 to 3

chunks_included:

- MATH10_CURRICULUM_OVERVIEW
- MATH10_CURRICULUM_STRUCTURE
- MATH10_UNIT_TOPIC_SUMMARY
- MATH10_UNIT1_ALGEBRA
- MATH10_UNIT1_TOPIC1_POLYNOMIALS
- MATH10_UNIT1_TOPIC2_LINEAR_EQUATIONS
- MATH10_UNIT1_TOPIC3_QUADRATIC_EQUATIONS
- MATH10_UNIT1_TOPIC4_ARITHMETIC_PROGRESSION
- MATH10_UNIT1_TOPIC5_RATIO_PROPORTION
- MATH10_UNIT2_COORDINATE_GEOMETRY
- MATH10_UNIT2_TOPIC1_COORDINATE_GEOMETRY
- MATH10_UNIT2_TOPIC2_GRAPHS
- MATH10_UNIT3_COMMERCIAL_MATHEMATICS
- MATH10_UNIT3_TOPIC1_BANKING_TAXATION

content_status: Structured

retrieval_status: Ingestion Ready

source_reference: "TP/CI-10/2026-27"

NEXT PART

Part 2 will include:

- Unit 4: त्रिकोणमिति
- त्रिकोणमितीय समीकरण एवं सर्वसमिकाएँ
- ऊँचाई एवं दूरी
- Unit 5: ज्यामिति
- ज्यामितीय आकृतियों में समरूपता
- वृत्त एवं स्पर्श रेखाएँ
- ज्यामितीय रचनाएँ
- Unit 6: गणितीय कथनों की जाँच

CHUNK 15: UNIT 4 — त्रिकोणमिति

chunk_id: MATH10_UNIT4_TRIGONOMETRY

chunk_type: subject_unit

unit_number: 4
unit_name: त्रिकोणमिति
unit_english_name: Trigonometry
marks: 10
periods: 30

विषय सामग्री

1. त्रिकोणमितीय समीकरण एवं सर्वसमिकाएँ

chunk_id: MATH10_UNIT4_TOPIC1_IDENTITYIES
topic_name: त्रिकोणमितीय समीकरण एवं सर्वसमिकाएँ
topic_english_name: Trigonometric Equations and Identities

मुख्य बिंदु:

- त्रिकोणमितीय अनुपातों के बीच संबंध
- एक अनुपात के पदों में अन्य अनुपातों को व्यक्त करना
- त्रिकोणमितीय सर्वसमिकाएँ
- त्रिकोणमितीय समीकरण एवं उनके हल
- पूर्ण कोणों के त्रिकोणमितीय अनुपात

प्रमुख सर्वसमिकाएँ:

$$\sin^2\theta + \cos^2\theta = 1$$

$$1 + \tan^2\theta = \sec^2\theta$$

$$1 + \cot^2\theta = \operatorname{cosec}^2\theta$$

Retrieval Keywords:

keywords:

- त्रिकोणमितीय अनुपात
- trigonometric ratios
- सर्वसमिका
- identity
- त्रिकोणमितीय समीकरण
- complete angles

2. ऊँचाई एवं दूरी

chunk_id: MATH10_UNIT4_TOPIC2_HEIGHT_DISTANCE
topic_name: ऊँचाई एवं दूरी
topic_english_name: Height and Distance

मुख्य बिंदु:

- उन्नयन कोण
- अवनमन कोण
- ऊँचाई एवं दूरी पर आधारित प्रश्न
- त्रिकोणमितीय अनुपातों का व्यावहारिक उपयोग

Retrieval Keywords:

keywords:

- ऊँचाई और दूरी
 - height and distance
 - उन्नयन कोण
 - angle of elevation
 - अवनमन कोण
 - angle of depression
-

CHUNK 16: UNIT 5 — ज्यामिति

chunk_id: MATH10_UNIT5_GEOMETRY

chunk_type: subject_unit

unit_number: 5

unit_name: ज्यामिति

unit_english_name: Geometry

marks: 12

periods: 30

विषय सामग्री

1. ज्यामितीय आकृतियों में समरूपता

chunk_id: MATH10_UNIT5_TOPIC1_SIMILARITY

topic_name: ज्यामितीय आकृतियों में समरूपता

topic_english_name: Similarity in Geometrical Shapes

मुख्य बिंदु:

- स्केलिंग
- आयत, वर्ग, वृत्त और त्रिभुज में समरूपता
- समरूपता की जाँच
- प्रमेय 1 से 8 तक

Retrieval Keywords:

keywords:

- समरूपता
 - similarity
 - scaling
 - समान त्रिभुज
 - geometrical shapes
-

2. वृत्त एवं स्पर्श रेखाएँ

chunk_id: MATH10_UNIT5_TOPIC2_CIRCLE_TANGENTS

topic_name: वृत्त एवं स्पर्श रेखाएँ

topic_english_name: Circle and Tangents

मुख्य बिंदु:

- जीवा
- चाप
- वृत्तखंड
- त्रिज्यखंड
- सर्वांगसम वृत्त
- केंद्र से जीवा पर लंब
- वृत्त पर आधारित प्रमेय 1 से 10
- स्पर्श रेखा
- छेदक रेखा
- स्पर्श रेखा एवं छेदक पर आधारित प्रमेय 11 से 14

Retrieval Keywords:

keywords:

- वृत्त
 - circle
 - जीवा
 - chord
 - चाप
 - arc
 - स्पर्श रेखा
 - tangent
 - छेदक रेखा
 - secant
-

3. ज्यामितीय रचनाएँ

chunk_id: MATH10_UNIT5_TOPIC3_CONSTRUCTIONS

topic_name: ज्यामितीय रचनाएँ

topic_english_name: Geometrical Constructions

मुख्य बिंदु:

- समरूप बहुभुज की रचना
- समरूप चतुर्भुज की रचना
- अंतर्गत वृत्त की रचना
- परिगत वृत्त की रचना

Retrieval Keywords:

keywords:

- ज्यामितीय रचना
 - geometrical construction
 - समरूप बहुभुज
 - similar polygon
 - अंतर्गत वृत्त
 - incircle
 - परिगत वृत्त
 - circumcircle
-

CHUNK 17: UNIT 6 — गणितीय कथनों की जाँच

chunk_id: MATH10_UNIT6_PROOF

chunk_type: subject_unit

unit_number: 6

unit_name: गणितीय कथनों की जाँच

unit_english_name: Proof of Mathematical Statements

marks: 4

periods: 10

विषय सामग्री

chunk_id: MATH10_UNIT6_TOPIC1_PROOF

topic_name: गणितीय कथनों की जाँच

topic_english_name: Proof of Mathematical Statements

मुख्य बिंदु:

- परिभाषा के आधार पर कथन सिद्ध करना

- पूर्व ज्ञात प्रमेयों का उपयोग
- अभिगृहीतों का उपयोग
- निगमनात्मक तर्कण
- गणितीय भाषा का प्रयोग
- गणितीय कथनों को सिद्ध करने की विधियाँ

Learning Outcomes:

- तार्किक क्रम में प्रमाण लिखना
- उचित परिभाषा और प्रमेय चुनना
- निष्कर्ष तक पहुँचने के चरण स्पष्ट करना
- गणितीय भाषा का शुद्ध उपयोग करना

Retrieval Keywords:

keywords:

- गणितीय प्रमाण
- mathematical proof
- गणितीय कथन
- deductive reasoning
- निगमनात्मक तर्कण
- प्रमेय
- theorem
- अभिगृहीत
- axiom

CHUNK 18: UNIT 7 — क्षेत्रमिति

chunk_id: MATH10_UNIT7_MENSURATION

chunk_type: subject_unit

unit_number: 7

unit_name: क्षेत्रमिति

unit_english_name: Mensuration

marks: 8

periods: 25

विषय सामग्री

ठोस आकृतियों का पृष्ठीय क्षेत्रफल एवं आयतन

chunk_id: MATH10_UNIT7_TOPIC1_SURFACE_AREA_VOLUME

topic_name: ठोस आकृतियों का पृष्ठीय क्षेत्रफल एवं आयतन

topic_english_name: Surface Area and Volume of Solids

मुख्य बिंदु:

- घन का पृष्ठीय जाल
- घनाभ का पृष्ठीय जाल
- घन एवं घनाभ के पृष्ठीय विकर्ण
- घन एवं घनाभ के आकाशीय विकर्ण
- बेलन का पृष्ठीय क्षेत्रफल एवं आयतन
- शंकु का पृष्ठीय क्षेत्रफल एवं आयतन
- गोले का पृष्ठीय क्षेत्रफल एवं आयतन

प्रमुख सूत्र:

Cube:

$$\text{Total Surface Area} = 6a^2$$

$$\text{Volume} = a^3$$

$$\text{Space Diagonal} = a\sqrt{3}$$

Cuboid:

$$\text{Total Surface Area} = 2(lb + bh + hl)$$

$$\text{Volume} = lbh$$

$$\text{Space Diagonal} = \sqrt{l^2 + b^2 + h^2}$$

Cylinder:

$$\text{Curved Surface Area} = 2\pi rh$$

$$\text{Total Surface Area} = 2\pi r(r + h)$$

$$\text{Volume} = \pi r^2 h$$

Cone:

$$\text{Slant Height} = \sqrt{r^2 + h^2}$$

$$\text{Curved Surface Area} = \pi rl$$

$$\text{Total Surface Area} = \pi r(l + r)$$

$$\text{Volume} = \frac{1}{3}\pi r^2 h$$

Sphere:

$$\text{Surface Area} = 4\pi r^2$$

$$\text{Volume} = \frac{4}{3}\pi r^3$$

Retrieval Keywords:

keywords:

- क्षेत्रमिति
- mensuration
- पृष्ठीय क्षेत्रफल
- surface area
- आयतन
- volume
- घन

- cube
 - घनाभ
 - cuboid
 - बेलन
 - cylinder
 - शंकु
 - cone
 - गोला
 - sphere
-

CHUNK 19: UNIT 8 — सांख्यिकी

chunk_id: MATH10_UNIT8_STATISTICS

chunk_type: subject_unit

unit_number: 8

unit_name: सांख्यिकी

unit_english_name: Statistics

marks: 6

periods: 15

विषय सामग्री

आँकड़ों का विश्लेषण

chunk_id: MATH10_UNIT8_TOPIC1_DATA_ANALYSIS

topic_name: आँकड़ों का विश्लेषण

topic_english_name: Data Analysis

मुख्य बिंदु:

- आलेखों में प्रदर्शित आँकड़ों का विश्लेषण
- अंकगणितीय औसत
- माध्यिका
- बहुलक
- औसत, माध्यिका और बहुलक का उपयोग
- अंतर्वेशन
- बहिर्वेशन

प्रमुख अवधारणाएँ:

Arithmetic Mean = Sum of Observations / Number of Observations

Median:

The middle value of arranged data.

Mode:

The value occurring most frequently.

Interpolation:

Estimating a value within the available data range.

Extrapolation:

Estimating a value outside the available data range.

Learning Outcomes:

- आँकड़ों को पढ़ना और समझना
- आलेख से निष्कर्ष निकालना
- औसत, माध्यिका और बहुलक ज्ञात करना
- उपयुक्त केंद्रीय प्रवृत्ति चुनना
- अंतर्वेशन और बहिर्वेशन का उपयोग करना

Retrieval Keywords:

keywords:

- सांख्यिकी
 - statistics
 - आँकड़ों का विश्लेषण
 - data analysis
 - अंकगणितीय औसत
 - arithmetic mean
 - माध्यिका
 - median
 - बहुलक
 - mode
 - अंतर्वेशन
 - interpolation
 - बहिर्वेशन
 - extrapolation
-

CHUNK 20: PROJECT WORK

chunk_id: MATH10_PROJECT_WORK

chunk_type: internal_assessment

section: Appendix A-01

total_marks: 25

duration: 2 Hours

periods: 20

मूल्यांकन योजना

Component	Marks
एक क्रियाकलाप	5
दो प्रोजेक्ट	10
अभिलेख एवं सत्रगत गतिविधि	5
मौखिक परीक्षा	5
Total	25

Project Requirements

विद्यार्थी को प्रायोजना सूची के प्रत्येक निर्धारित खण्ड से दो-दो प्रायोजनाओं का अभिलेख संधारित करना अनिवार्य है।

Project areas include:

- बीजगणित
- ज्यामिति
- त्रिकोणमिति

Recommended evidence:

- Activity record
- Calculations
- Diagrams
- Observations
- Conclusion
- Viva responses

Retrieval Keywords:

keywords:

- प्रायोजना कार्य
- mathematics project
- activity
- record work
- sessional activity
- viva

CHUNK 21: QUESTION PAPER DISTRIBUTION

chunk_id: MATH10_QUESTION_DISTRIBUTION

chunk_type: examination_scheme

section: Appendix B

theory_marks: 75

duration: 3 Hours

Question-Type Distribution

Question Type	Marks per Question	Number of Questions	Total Marks
Objective / One-Word Answer	1	15	15
Very Short Answer	2	5	10
Short Answer	3	5	15
Long Answer-I	4	5	20
Long Answer-II	5	3	15
Total		19	75

Objective Question Structure

Question 1 contains two sections:

section_a:

type: Multiple Choice Questions

abbreviation: MCQ

questions: 10

marks_each: 1

section_b:

type: One Word Answer

abbreviation: OWA

questions: 5
marks_each: 1

CHUNK 22: QUESTION PAPER STRUCTURE

chunk_id: MATH10_QUESTION_PAPER_STRUCTURE
chunk_type: question_paper_pattern
section: Appendix C-01

Question Numbering

question_1:
type:
- MCQ
- One Word Answer
total_marks: 15

questions_2_to_6:
type: Very Short Answer
marks_each: 2
total_marks: 10

questions_7_to_11:
type: Short Answer
marks_each: 3
total_marks: 15

questions_12_to_16:
type: Long Answer-I
marks_each: 4
total_marks: 20

questions_17_to_19:
type: Long Answer-II
marks_each: 5
total_marks: 15

Internal Choice

तारांकित प्रश्नों में आंतरिक विकल्प दिए जाएंगे।

Units containing starred questions include:

- बीजगणित
- वाणिज्य गणित
- त्रिकोणमिति
- ज्यामिति
- क्षेत्रमिति
- सांख्यिकी

CGBSE CLASS 10 MATHEMATICS CURRICULUM

Academic Session 2026–27

Ingestion-Ready Structured Version — Part 4

CHUNK 23: COGNITIVE LEVEL DISTRIBUTION

chunk_id: MATH10_COGNITIVE_DISTRIBUTION

chunk_type: assessment_framework

section: Appendix B

theory_marks: 75

Learning Objective Distribution

Cognitive Level	Marks	Weightage
Knowledge	15	20%
Understanding	19	25%
Application	18	25%

Analysis (HOTS)	8	10%
Evaluation	8	10%
Creativity	7	10%
Total	75	100%

Assessment Focus

- **Knowledge:** Definitions, facts, formulas, recall
- **Understanding:** Concepts, explanation, interpretation
- **Application:** Solving real-world mathematical problems
- **Analysis:** Logical reasoning, comparison, HOTS questions
- **Evaluation:** Justification, decision making, mathematical reasoning
- **Creativity:** Planning, modelling and mathematical thinking

Retrieval Keywords

keywords:

- Bloom taxonomy
- cognitive level
- knowledge
- understanding
- application
- analysis
- evaluation
- creativity

CHUNK 24: UNIT-WISE BLUEPRINT

chunk_id: MATH10_BLUEPRINT

chunk_type: blueprint

section: Appendix C

Marks Distribution by Unit

Unit	Marks
Algebra	20
Coordinate Geometry	8
Commercial Mathematics	7

Trigonometry	10
Geometry	12
Proof of Mathematical Statements	4
Mensuration	8
Statistics	6
Total	75

Blueprint Highlights

- 15 Objective Questions (MCQ + OWA)
- 5 Very Short Answer Questions
- 5 Short Answer Questions
- 5 Long Answer-I Questions
- 3 Long Answer-II Questions
- Internal choice in starred questions

Retrieval Keywords

keywords:

- blueprint
 - unit weightage
 - marks distribution
 - exam blueprint
-

CHUNK 25: DIFFICULTY LEVEL DISTRIBUTION

chunk_id: MATH10_DIFFICULTY_LEVEL

chunk_type: difficulty_distribution

section: Examination Pattern

Difficulty Classification

The Mathematics examination follows a balanced competency-based assessment model.

Level	Purpose
Easy	Basic concepts, definitions, direct formula application
Moderate	Conceptual understanding and multi-step problems
Difficult	Higher Order Thinking Skills (HOTS), analytical reasoning
Assessment aims to evaluate:	
• Conceptual clarity • Mathematical reasoning • Computational accuracy • Logical proof • Real-life application • Problem-solving ability	
Retrieval Keywords	
keywords: - easy questions - moderate questions - difficult questions - HOTS - competency	

CHUNK 26: MATHEMATICAL COMPETENCIES

chunk_id: MATH10_LEARNING_COMPETENCIES
 chunk_type: competency_framework
 section: Learning Outcomes

Expected Student Competencies

Students should be able to:

- Solve algebraic equations
- Apply trigonometric identities
- Construct geometric figures
- Prove mathematical statements
- Interpret graphs and coordinate systems
- Calculate surface area and volume

- Analyze statistical data
 - Apply mathematics to banking and taxation
 - Solve real-life mathematical problems
 - Communicate mathematical reasoning logically
-

Core Skills

skills:

- Problem Solving
 - Critical Thinking
 - Logical Reasoning
 - Mathematical Communication
 - Analytical Skills
 - Data Interpretation
 - Financial Literacy
-

CHUNK 27: RAG METADATA SCHEMA

chunk_id: MATH10_RAG_SCHEMA
chunk_type: metadata_schema
section: Vector Database

Recommended Metadata

board: CGBSE
class: 10
subject: Mathematics
subject_hindi: गणित

unit:
topic:
subtopic:

marks:
periods:

difficulty:

question_type:

learning_outcomes:

keywords:

formulae:

language: Hindi

academic_year: 2026-27

document_type: Curriculum

chunk_id:

source_reference:

Search Filters

- Unit
- Topic
- Formula
- Question Type
- Marks
- Difficulty
- Learning Outcome
- Chapter
- Academic Year

- Retrieval Guidelines

chunk_range:

start: MATH10_COGNITIVE_DISTRIBUTION

end: MATH10_RAG_GUIDELINES

document_status: Ingestion Ready

optimized_for:

- RAG
- Semantic Search
- Educational Chatbots
- AI Tutor
- Vector Database
- Fine Tuning

source_reference: TP/CI-10/2026-27

CGBSE CLASS 10 MATHEMATICS CURRICULUM

Academic Session 2026–27

Ingestion-Ready Structured Version — Part 5

CHUNK 29: FORMULA INDEX

chunk_id: MATH10_FORMULA_INDEX

chunk_type: formula_index

section: Formula Repository

Major Formula Categories

Algebra

- Polynomial Identities
- Quadratic Formula
- Sum & Product of Roots
- Arithmetic Progression Formula

Coordinate Geometry

- Distance Formula
- Slope Formula
- Equation of Line

Trigonometry

- Trigonometric Ratios
- Standard Identities
- Height & Distance

Geometry

- Similarity Theorems
- Circle Theorems
- Tangent Properties

Mensuration

- Cube
- Cuboid
- Cylinder
- Cone
- Sphere

Statistics

- Arithmetic Mean
 - Median
 - Mode
-

CHUNK 30: MATHEMATICAL ENTITY INDEX

chunk_id: MATH10_ENTITY_INDEX

chunk_type: entity_dictionary

section: Knowledge Graph

Entity Categories

Concepts

- Polynomial
- Equation
- Ratio
- Progression
- Coordinate
- Slope
- Triangle
- Circle
- Tangent
- Cone
- Cylinder
- Sphere
- Mean
- Median
- Mode

Mathematical Objects

- Point
- Line
- Angle
- Graph
- Curve
- Radius
- Diameter
- Chord
- Arc

Skills

- Simplification
- Construction
- Proof
- Interpretation
- Calculation
- Verification
- Problem Solving

CHUNK 31: SEARCH ALIAS DICTIONARY

chunk_id: MATH10_ALIAS_DICTIONARY

chunk_type: alias_mapping

section: Search Optimization

Hindi ↔ English Mapping

बहुपद: Polynomial

द्विघात: Quadratic

समांतर श्रेणी: Arithmetic Progression

निर्देशांक: Coordinate

आलेख: Graph

त्रिकोणमिति: Trigonometry

समरूपता: Similarity

वृत्त: Circle

स्पर्श रेखा: Tangent

क्षेत्रमिति: Mensuration

सांख्यिकी: Statistics

माध्यिका: Median

बहुलक: Mode

औसत: Mean

CHUNK 32: QUESTION TAGGING SCHEMA

chunk_id: MATH10_QUESTION_TAGS
chunk_type: metadata_tags
section: Assessment

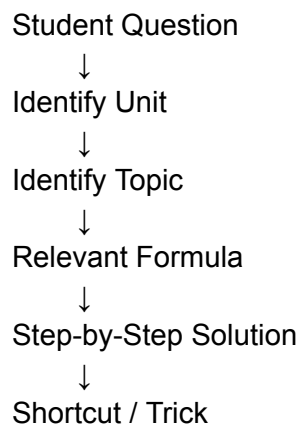
Each generated question should include:

unit:
topic:
difficulty:
marks:
question_type:
formula_used:
concept:
learning_outcome:
exam_frequency:
board_exam_relevant:

CHUNK 33: AI TUTOR RESPONSE TEMPLATE

chunk_id: MATH10_AI_RESPONSE_TEMPLATE
chunk_type: response_template
section: AI Tutor

Recommended Response Flow



↓
Common Mistakes
↓
Exam Tip
↓
Practice Question

Every mathematical response should include:

- Concept
 - Formula
 - Solution Steps
 - Final Answer
 - Verification
 - Related Topic
-

CHUNK 34: QUALITY ASSURANCE

chunk_id: MATH10_QA_CHECKLIST

chunk_type: validation

section: Production QA

Validation Checklist

- ✓ Unicode Verified
- ✓ Mathematical Symbols Verified
- ✓ Formula Formatting Correct
- ✓ Topic Mapping Complete
- ✓ Marks Verified
- ✓ Blueprint Verified
- ✓ Chunk IDs Unique
- ✓ Search Keywords Added
- ✓ Metadata Complete
- ✓ RAG Tested

✓ AI Tutor Ready

CHUNK 35: PRODUCTION METADATA

chunk_id: MATH10_PRODUCTION_METADATA

chunk_type: deployment

section: Production

document_name: CGBSE Class 10 Mathematics

version: 1.0

curriculum: 2026-27

board: CGBSE

language: Hindi

subject: Mathematics

status: Production Ready

optimized_for:

- RAG
- Educational AI
- Semantic Search
- Hybrid Search
- AI Tutor
- Question Generation
- Formula Retrieval
- Educational Chatbot
- Vector Database

COMPLETE DOCUMENT SUMMARY

total_parts: 5

total_chunks: 35

contains:

- Curriculum Structure
- Unit-wise Syllabus
- Detailed Topics
- Project Work
- Blueprint
- Question Paper Pattern
- Cognitive Distribution
- Formula Index
- Entity Dictionary
- Search Aliases
- Question Metadata
- AI Tutor Template
- QA Checklist
- Production Metadata

deployment_ready: true

END OF DOCUMENT

END OF MATHEMATICS CURRICULUM